



Artificial Intelligence: An introduction

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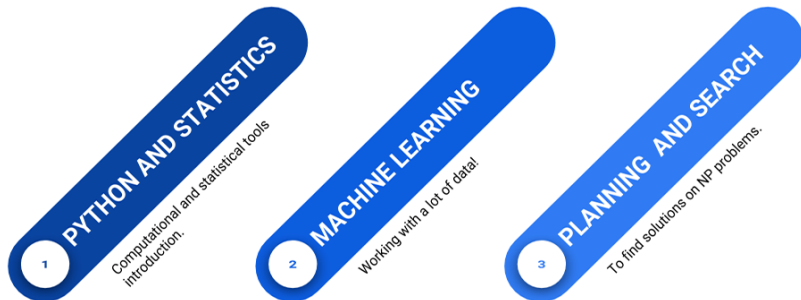


Biomedical Imaging, Vision and Learning Laboratory

Información del profesor

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- ▶ Contenido del curso:
<https://gitlab.com/bivl2ab/academico/cursos-uis/ai/ai-uis-student>

Course outline



- ▶ **The course** is mainly based on the **coding of I.A concepts** and the **development of small applications**
- ▶ **The course** is focused on the analysis and study of **I.A algorithms**

I.A. Course



Course

It is about learning the fundamental tools to face problems and challenges of I.A. as well as some computational tools and theoretical concepts.

Second Module: Machine Learning

- ▶ Naive Bayes
- ▶ Classification trees
- ▶ Support Vector Machines
- ▶ Non-supervised learning
- ▶ Introduction to Deep-learning



Third module: Planning and Searching

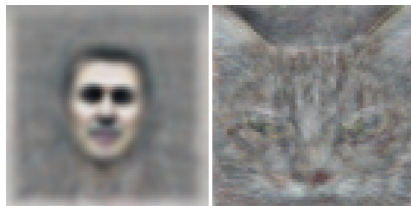
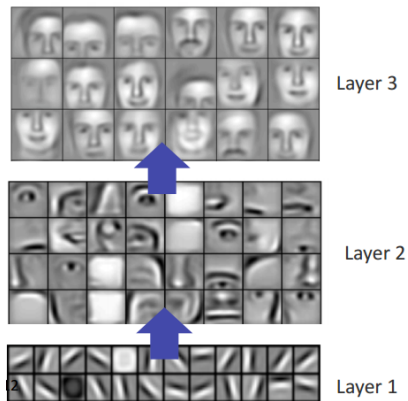
- ▶ Planning and searching
- ▶ Genetic algorithms (GA)
- ▶ Simulated annealing (SA)



Big challenges

Now, algorithms can create new things ... as abstract as the art! (Netflix 5:15, Bill Nye)

Big challenges



Big challenges

Video 01



ALPHAGO

Big challenges

Video 02



ALPHAGO

ImageNet Challenge

IMAGENET

IMAGENET Netflix 22:30, Bill Nye

- 1,000 object classes (categories).
- Images:
 - 1.2 M train
 - 100k test.

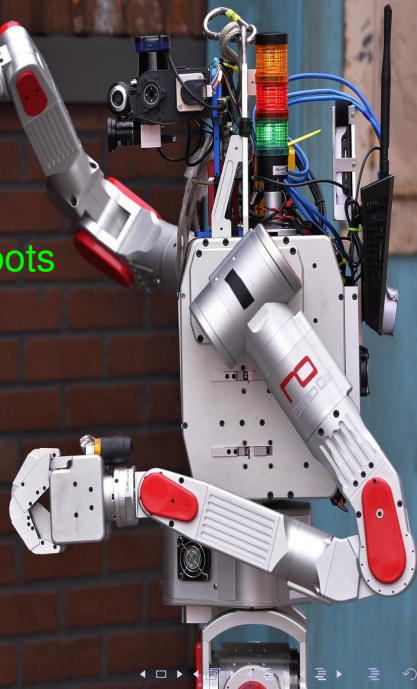


As support for decisions making

- ▶ Medicine
- ▶ Politics
- ▶ Laws



NOVA Rise of the Robots





y ahora ...Atlas. Google again!

Boston Dynamics

DARPA



Structure of the Course

- ▶ 40% Talleres (Problemsets)
- ▶ 30% Parciales (Quizes)
- ▶ 30% + [10% ,20 %, 30 %] Proyecto funcional IA
 - ▶ 10%: Hasta + una unidad en una nota de un 10% (los parciales por ejemplo) o su equivalente en otros porcentajes.
 - ▶ Las notas o porcentajes adicionales se obtienen si fueron puntuales en todas las entregas.
- ▶ +10% Online courses (MOOC)
 - ▶ Esta vez, unicamente se aceptan cursos relacionados con I.A., machine learning, análisis de datos, o tecnologías relacionadas.
 - ▶ Solo se evalua el curso, si ha sido previamente aprobado antes del primer corte.

Curso en GITLAB:

<https://gitlab.com/bivl2ab/academico/cursos-uis/ai/ai-uis-student>

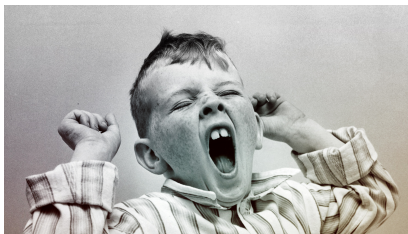
PROJECTS 2019-1

Proyectos realizados en 2019-1:

`https://gitlab.com/bivl2ab/academico/
cursos-uis/ai/ai-uis-student/blob/master/
projects/Proyectos_2019-1.md`



Thank you for your attention ...



... It's time to wake up

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